

Bloom filters

- * A memory efficient data-structure used to test whether an element might be in a set.
- * It can tell you if something is definitely NOT present.
or if something is possibly present.
- * If bloom filter says :- 'Not present' $\rightarrow$ 100% sure
(0% chance of it being there)

Key Idea

- Bloom filters act like fast pre-check layer.
- widely used in large scale systems to reduce database lookups, disk reads & network calls.

Working (conceptual)

- Bloom filter uses :
 - A fixed-size bit array
 - Multiple hash functions
- When inserting an item:-
 - Hash it multiple times
 - Set corresponding bits to 1.
- When checking an item :
 - Hash it in the same way.
 - Check those bit positions.
- If any bit is 0 $\rightarrow$ definitely not present
- If all bits are 1 $\rightarrow$ possibly present.

Location-based Database

* Location queries are different from normal queries, they include:-

- Latitude
- Longitude
- Distance calculation
- Radius search
- Nearest neighbour search.

• A database that efficiently stores & queries geographical coordinates and spatial data. Supports all above points.

• They use special data structures like:-

- R-Tree
- Geohashes
- Quadrees
- Spatial indexes

→ These structures divide the world into regions.

Instead of scanning entire db, table:-

- Looks only in relevant geographic region
- Quickly filters nearby points

→ This makes 'near me' queries fast (Google maps)

Popular Options

- PostgreSQL with PostGIS
- Elasticsearch (geo queries)
- MongoDB (geospatial index)
- Redis (geo commands)

* Used in Ride-sharing, food delivery, store locator, track shipments, geo-fencing.

Ex:- Every driver updates location every few seconds.

This creates:-

- High write load
- High read load.

- R-Tree → Groups nearby objects into rectangles
- Geo hashes → Converts longitude + latitude into a short string.
- Quad tree → Divides spaces into four parts repeatedly.
 - Top left
 - Top right
 - Bottom left
 - Bottom right(Then again each region)